Scientific Experimentation and Evaluation

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Assignment 1: Measurement System Design

I. Robot and Measurement System

The robot used in this experiment is a **LEGO EV3 differential drive robot**, consisting of two main wheels on the same axle and a small supporting caster at the front. The task of the experiment is to measure the stop position of the robot after executing a predefined motion.

In the initial design, a **lever and pencil nib mechanism** was used to mark the stop position. However, it was later replaced with a **pen refill-based marking system**, which makes the system more robust for rough terrain. A pen refill is directly mounted on both sides of the robot, allowing it to make a mark on the cardboard surface. This modification simplifies the setup and improves marking consistency.

- The midpoint between the two pen marks corresponds approximately to the final position of the robot's wheel axle center.
- The orientation θ can be estimated from the line connecting the left and right marks, since they reflect the alignment of the robot's axle.

Reason for Two-Point Measurement: Two points are sufficient to define the robot's axle line and its midpoint, which directly correspond to the robot's final pose on the grid. Introducing a third point would substantially increase the measurement and data processing. Performing 25 trials for each of the three motion types and using three markers would expand this to 225 individual coordinate readings. Additionally, computing the centroid of a triangular configuration manually would complicate data handling and introduce more room for error. Therefore, the two-point setup achieves the necessary accuracy while keeping the experimental process straightforward and repeatable. The midpoint position and the orientation line ensure reliable results with minimal hardware.

II. Measurement Process

The process for carrying out the measurements is as follows:

1) A large cardboard sheet with a printed or drawn grid is fixed to the working surface using tape, so it does not move during the experiment.
2) A clear origin is marked on the grid, including both position and orientation. To ensure repeatability, a simple placement template is used so the robot always starts from the same position and orientation.
3) The robot is placed at the start position and the program for the chosen motion (straight, left arc, or right arc) is executed.
4) When the robot comes to a stop, the pen refills on both sides can be rotated 90° to avoid the stopper and pushed down simply to mark the spot, as shown in Figure 2. If both marks are recorded, their midpoint represents the final position (x, y). The line through the marks gives the orientation θ.
5) The (x, y, θ) values are measured relative to the origin on the grid and recorded in a results table.
6) The robot is reset to the origin and the process is repeated at least 25 times for each type of motion.

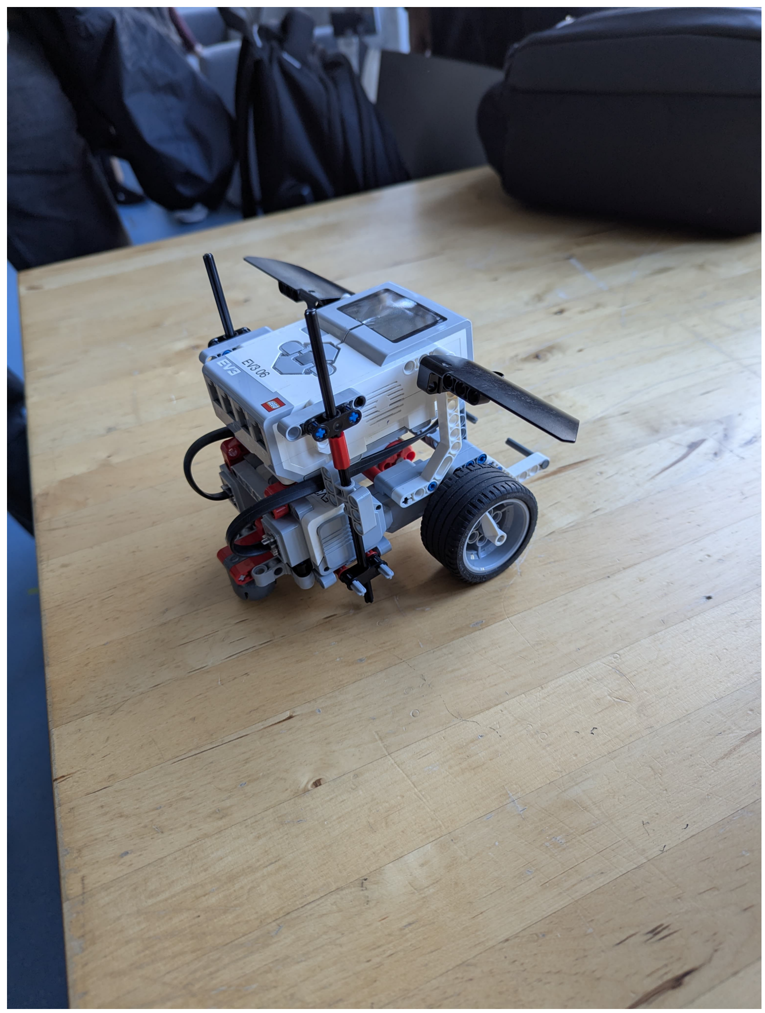

Fig. 1: Robot with lever system.

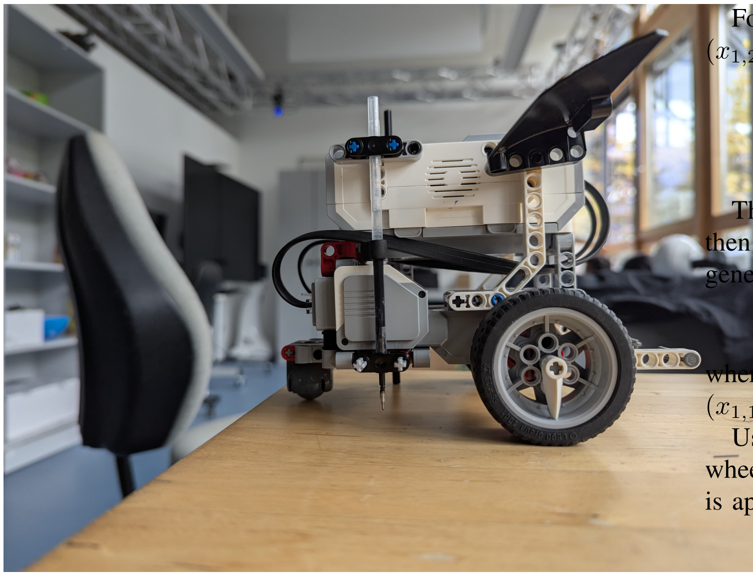

Fig. 2: Robot with modified pen refill marking mechanism.

III. Expected Problems and Mitigations

During the experiment, several practical issues may occur:

- **Ink spreading or uneven marking:** Refills may release excess ink or create faint marks. **Solution:** Use refills with consistent ink flow and test before trials.
- **Mounting stability:** If the refill is loosely mounted, mark positions may vary. **Solution:** Tape the refill to reduce any gap and improve stability.
- **Start placement variation:** Inconsistent starting positions can add systematic errors. **Solution:** Use a fixed template or taped outline for robot placement.
- **Cardboard movement:** If the grid sheet shifts, all measurements are affected. **Solution:** Secure the sheet with tape to the surface.

IV. Jacobian Error Propagation

The propagation of uncertainty in the measured input coordinates requires the calculation of the Jacobian matrix $\mathbf{J}$. The total input vector $\mathbf{x} = (x_{1,1}, x_{1,2}, x_{2,1}, x_{2,2})^T$ is composed of all four scalar variables representing the coordinates of the two pen marks. Since $\mathbf{F}$ is a vector-valued function with three outputs (final pose: x, y, θ) and four inputs, the resulting Jacobian matrix $\mathbf{J}$ is a 3×4 matrix.

For the orientation θ, let the measured coordinates be

$$\mathbf{p} = \begin{bmatrix} x_L & y_L & x_R & y_R \end{bmatrix}^\top,$$

where the robot orientation is computed as

$$\theta(\mathbf{p}) = \operatorname{atan2}(y_R - y_L, x_R - x_L).$$

The Jacobian for orientation is given by:

$$\mathbf{J}_\theta = \frac{\partial \mathbf{F}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial F_1}{\partial x_{1,1}} & \frac{\partial F_1}{\partial x_{1,2}} & \frac{\partial F_1}{\partial x_{2,1}} & \frac{\partial F_1}{\partial x_{2,2}} \\ \frac{\partial F_2}{\partial x_{1,1}} & \frac{\partial F_2}{\partial x_{1,2}} & \frac{\partial F_2}{\partial x_{2,1}} & \frac{\partial F_2}{\partial x_{2,2}} \\ \frac{\partial F_3}{\partial x_{1,1}} & \frac{\partial F_3}{\partial x_{1,2}} & \frac{\partial F_3}{\partial x_{2,1}} & \frac{\partial F_3}{\partial x_{2,2}} \end{pmatrix}$$

For the third row (orientation), with $d = (x_{1,1} - x_{2,1})^2 + (x_{1,2} - x_{2,2})^2$:

$$= \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{x_{1,2}-x_{2,2}}{d} & \frac{x_{2,1}-x_{1,1}}{d} & \frac{x_{2,2}-x_{1,2}}{d} & \frac{x_{1,1}-x_{2,1}}{d} \end{pmatrix}$$

The covariance matrix of the calculated end pose $\mathbf{C}_F$ is then determined from the input covariance matrix $\mathbf{C}_x$ via the generalized error propagation formula:

$$\mathbf{C}_F = \mathbf{J}\mathbf{C}_x\mathbf{J}^T, \tag{1}$$

where $\mathbf{C}_x$ is the 4×4 covariance matrix of the input variables $(x_{1,1}, x_{1,2}, x_{2,1}, x_{2,2})$.

Using the reported position uncertainty $\sigma_{\text{pos}} \approx 0.36$ mm and wheel-axle separation $L \approx 96$ mm, the orientation uncertainty is approximately:

$$\boxed{\sigma_\theta \approx 0.0053 \text{ rad} \approx 0.30^\circ}$$

V. Conclusion

This measurement system provides a simple, reliable, and precise method for determining the robot's final position and orientation. The direct contact of the refill with the surface ensures consistent marking with minimal error, reducing variability in results. Its compact design minimizes mechanical complexity and allows easy alignment on both sides of the robot. Due to the fine tip of the pen refill, the marking precision is high, resulting in a position uncertainty of approximately ± 0.4 mm and an orientation uncertainty of about $\pm 0.2^\circ$.

Among the possible configurations, the two-point method proved to be the most straightforward and practical choice, as it provides all required position and orientation information without unnecessary complexity or excessive data recording.

Assignment 2: Manual Motion Observation

I. Introduction

This report presents the execution and results of Assignment 2, where we conducted manual measurements of our LEGO EV3 differential drive robot's end poses across three different motion types: straight, left arc, and right arc. Each motion was repeated 25 times to gather statistical data on the robot's behavior and positioning accuracy.

II. Program and Parameters

The robot was controlled using the provided EV3 control program with the following predefined parameters:

- **Wheel diameter:** 5.30 cm
- **Distance between wheels:** 12.00 cm
- **Pen refill offset X:** 5.50 cm
- **Pen refill offset Y:** 6.40 cm
- **Motion types:** Left arc, Straight line, Right arc
- **Trials per motion:** 25 trials each (75 total)

III. Observations During Execution

During the experimental runs, the following observations were made:

1) **Surface consistency:** The cardboard surface remained stable throughout all trials, providing a reliable measurement platform.
2) **Marking precision:** The pen refill mechanism produced consistent marks with minimal ink spreading. The fine tip allowed for precise position recording.
3) **Start position repeatability:** Using a fixed template ensured consistent starting positions across all trials, minimizing systematic errors.
4) **Robot behavior variations:** Slight variations in end positions were observed, likely due to:
   - Motor encoder resolution limitations
   - Battery voltage variations affecting motor performance
   - Minor inconsistencies in floor levelness
5) **Motion characteristics:**
   - *Straight motion:* Generally consistent forward movement.
   - *Left arc:* Smooth curved trajectory with good repeatability
   - *Right arc:* Similar behavior to left arc, symmetric results observed

IV. Data Collection and Transformation

The experiment was conducted on a large cardboard sheet with a printed grid, which served as the global coordinate frame. A fixed starting pose was used for all trials to ensure consistency.

For each of the 25 runs per motion (left arc, straight, right arc), the following procedure was followed:

- The robot was placed at the designated origin (0, 0).
- The corresponding motion program was executed.
- Upon stopping, the two pen refills were used to mark the robot's position.
- The coordinates of the two marks, $(X1, Y1)$ and $(X2, Y2)$, were recorded.

The raw marker coordinates were then transformed to calculate the final pose of the wheel axle center. This involved a three-step mathematical process:

1) **Midpoint Calculation:** The midpoint of the pen refills, (Avg X, Avg Y), was found by averaging the coordinates:
$$\text{X} = \frac{X1 + X2}{2}, \quad \text{Y} = \frac{Y1 + Y2}{2}$$
2) **Orientation Calculation:** The final orientation, θ, was calculated based on the position of the robot's axle center. By convention, an orientation of $\theta = 0$ corresponds to the robot facing along the positive Y-axis.
The radius of curvature is:
$$r = \frac{X^2 + Y^2}{2X}$$
The orientation angle θ is defined as:
$$\theta = \arctan \frac{Y2 - Y1}{X2 - X1}$$
Finally, the orientation is rounded to two decimal places:
$$\theta_{\text{final}} = \text{round}(\theta, 2)$$

V. Data Visualization

The collected data has been visualized in three comprehensive plots showing the robot's behavior for each motion type. The visualizations include both the manually measured end poses and the complete paths recorded by the encoder data.

A. Manual Markings on Paper

Figure 3 presents the markings of robot position.

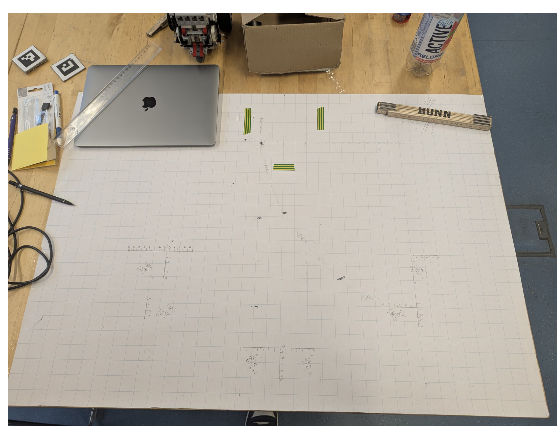

Fig. 3: Manual Measurements and markings for each run of robot

B. End Poses Visualization

Figure 4 shows the distribution of the robot's end poses for all three motion types from the manual measurements. This plot illustrates the spread and clustering of final positions, providing insight into the precision of the robot's motion control.

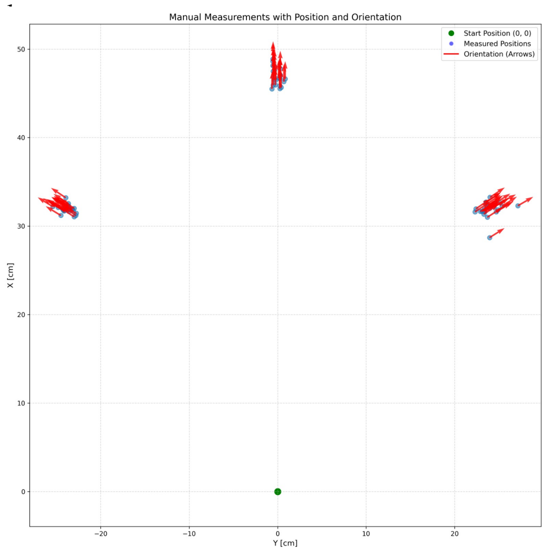

Fig. 4: Robot end poses from manual measurements for straight, left arc, and right arc motions. Each point represents the compensated midpoint position between the two pen marks after refill offset correction.

C. Complete Robot Paths

Figure 5 displays the complete trajectories of the robot as recorded by the encoder measurements. This visualization shows the continuous path taken by the robot from start to finish for all trials.

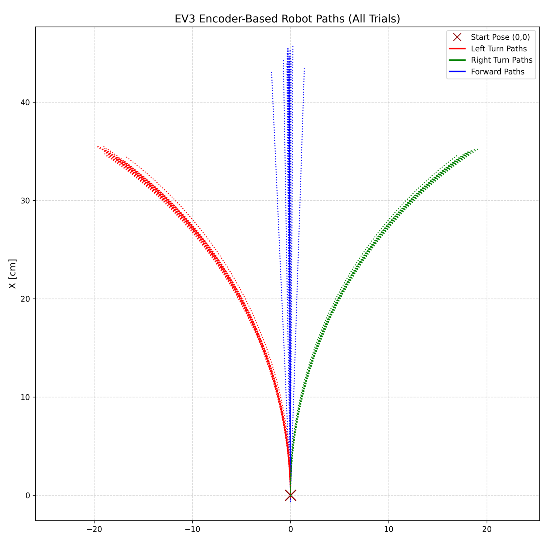

Fig. 5: Complete robot paths from encoder measurements showing the trajectories for all three motion types. The paths demonstrate the robot's movement behavior from the starting pose to the final positions.

D. Combined Visualization

Figure 6 presents a combined view that overlays the manually measured end poses with the encoder-based path data, allowing for direct comparison between the two measurement methods.

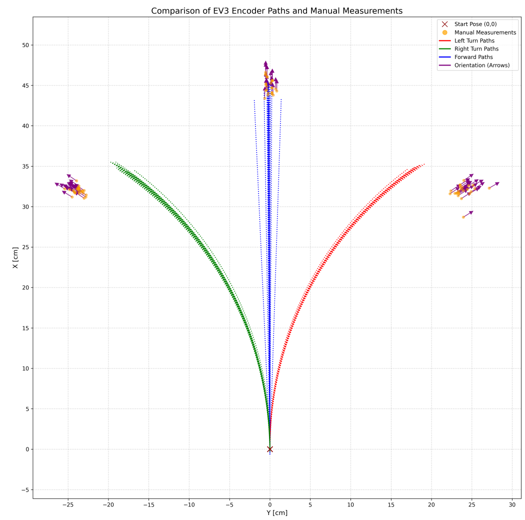

Fig. 6: Combined visualization showing both encoder-based paths and manually measured end poses. This allows for comparison between the internal sensor data and external measurements.

E. Combined Visualization of All Groups

Figure 7 presents a combined view of data taken from other groups.

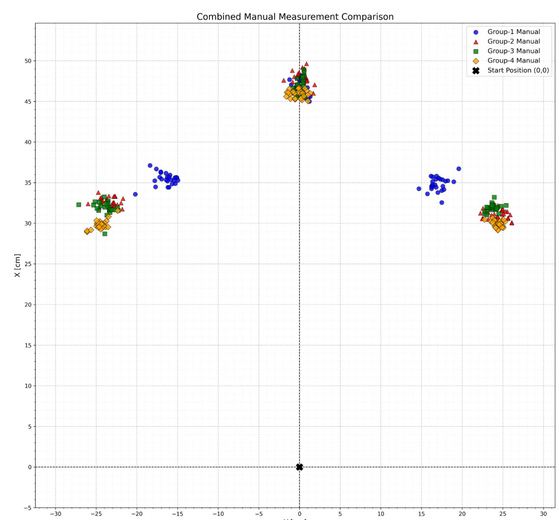

Fig. 7: Combined visualization showing manually measured end poses from other groups and comparing with our group.

F. Combined Visualization of Trajectories

Figure 8 presents a combined view of data taken from other groups.

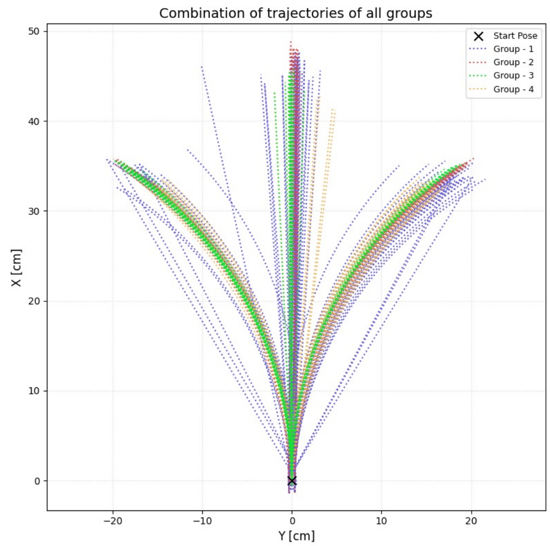

Fig. 8: Combined visualization showing encoder data from other groups and comparing with our group.

VI. Manual Measurement Data

TABLE I: Manual Measurements for Straight, Left Arc, and Right Arc Motions (All values in cm, orientation in degrees)

Straight Motion				**Left Arc Motion**				**Right Arc Motion**			
Trial	**X**	**Y**	θ	**Trial**	**X**	**Y**	θ	**Trial**	**X**	**Y**	θ
1	47.03	0.30	0.00	1	31.45	-22.75	-62.10	1	32.65	25.15	62.61
2	45.87	0.25	-0.59	2	31.65	-23.60	-62.59	2	31.90	23.35	56.48
3	45.54	0.25	-2.36	3	31.85	-23.65	-64.29	3	31.60	22.30	55.98
4	47.01	0.25	-1.18	4	31.20	-22.80	-55.98	4	31.00	23.70	56.98
5	45.67	0.40	-1.19	5	31.05	-23.00	-56.65	5	31.60	23.40	56.98
6	45.94	-0.25	-1.18	6	31.90	-23.40	-56.98	6	31.95	22.40	55.65
7	47.61	-0.35	-0.59	7	31.20	-24.50	-58.63	7	31.70	23.35	57.48
8	47.50	-0.50	0.00	8	31.85	-23.25	-57.15	8	31.35	23.30	56.31
9	47.25	-0.40	0.60	9	31.70	-23.10	-55.98	9	31.70	23.20	59.04
10	46.52	-0.20	-0.60	10	31.95	-23.20	-57.30	10	32.30	27.15	60.00
11	46.71	0.25	-1.18	11	32.20	-23.90	-56.98	11	32.00	23.95	66.10
12	48.88	-0.55	1.21	12	32.20	-23.65	-60.18	12	33.25	24.00	56.31
13	48.61	-0.55	1.77	13	32.30	-23.65	-56.48	13	32.60	24.60	57.62
14	46.73	0.10	0.00	14	32.35	-23.75	-62.57	14	28.70	23.95	57.80
15	46.50	-0.40	-0.61	15	32.10	-24.80	-58.63	15	31.65	22.90	56.31
16	46.63	0.85	0.00	16	32.00	-24.35	-58.74	16	32.20	23.65	56.48
17	45.93	-0.50	0.60	17	31.75	-24.10	-58.31	17	31.65	23.10	58.31
18	46.84	-0.50	1.19	18	31.70	-24.15	-58.12	18	32.70	23.55	56.48
19	45.50	-0.65	1.21	19	32.55	-23.65	-56.80	19	32.70	23.55	56.48
20	47.81	0.30	0.00	20	32.10	-23.75	-56.48	20	32.15	24.45	59.44
21	46.32	0.70	0.58	21	32.20	-25.40	-58.63	21	32.45	23.55	56.80
22	48.14	-0.55	1.77	22	32.20	-23.65	-60.18	22	32.25	25.35	57.53
23	48.01	-0.35	-0.59	23	33.20	-23.95	-56.14	23	31.60	24.70	58.63
24	46.67	0.25	-2.36	24	31.85	-24.25	-57.80	24	31.85	24.90	61.66
25	47.06	0.30	0.00	25	32.00	-23.00	-55.30	25	32.40	23.80	57.62

VII. Comparison with Expectations

The observed behavior of the robot generally matches our expectations from Assignment 1:

1) **Motion repeatability:** All three motion types show relatively tight clustering of end poses, indicating good repeatability of the robot's motion control.
2) **Systematic patterns:** The straight motion shows minimal lateral drift (mean X 0), while the left and right arcs show symmetric behavior with respect to the Y-axis, as expected.
3) **Variation sources:** The spread in end positions aligns with anticipated error sources including encoder resolution, wheel slippage, and battery variations.

VIII. Data Transformation and Combined Dataset

Our data follows the standard coordinate convention agreed upon by all groups:

- X-axis aligned with forward robot direction
- Y-axis perpendicular (right-hand rule)
- All positions transformed to represent the robot's axle center
- All values rounded to 2 decimal places

IX. Conclusion

The experiment successfully captured 75 data points (25 trials × 3 motion types) documenting the robot's end pose behavior. The pen refill marking mechanism proved reliable and precise, and the observed motion patterns demonstrate good repeatability. The visualizations clearly show the distribution of end poses and provide insight into the robot's motion characteristics.

The comparison between manually measured end poses and encoder-based trajectories reveals the strengths and limitations of both measurement approaches. The manual measurements provide ground truth data for the robot's final position, while the encoder data captures the complete motion trajectory.

X. Appendix: Video Documentation

Three video recordings were captured showing the robot's behavior during the experiment:

- `straight_motion.mp4` - Robot executing straight line motion
- `left_motion.mp4` - Robot executing left motion
- `right_arc_motion.mp4` - Robot executing right arc motion

Assignment 3: Statistical Evaluation of Robot Motion

I. Introduction

This report is an extension of the analysis from last Assignment, here we have performed a statistical evaluation of the robot's motion behavior for the three motion types: straight, left arc, and right arc. The primary goal is to characterize the observed motion in terms of statistical parameters and check whether the end poses of the robot follow a Gaussian distribution.

The assignment includes data preprocessing, Gaussian fitting, outlier removal, Principal Component Analysis (PCA), and comparison of uncertainties between manual and encoder-based measurements. All analyses are performed on the combined dataset from all groups.

II. Mean and Norm between manual end poses and final reading of encoder data

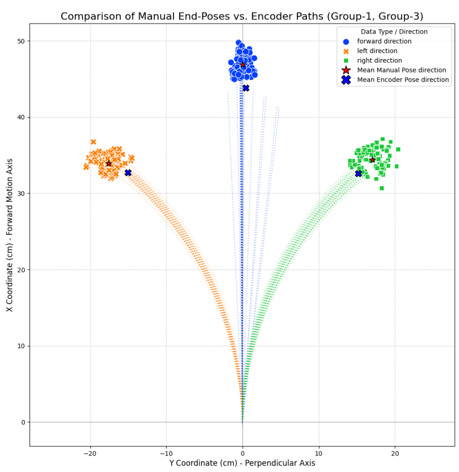

Fig. 9: Comparison of manual end poses and encoder data.

A. Accuracy Calculation (L2 Norm)

- Mean of the manual poses of forward: [-0.04, 46.89] cm
- Mean of the end poses from encoder data of forward: [0.45, 43.83] cm
- Accuracy Error (L2 Norm) for forward: 3.10 cm
- Mean of the manual poses of left: [-22.38, 31.93] cm
- Mean of the end poses from encoder data of left: [-15.08, 32.71] cm
- Accuracy Error (L2 Norm) for left: 7.35 cm
- Mean of the manual poses of right: [22.12, 32.45] cm
- Mean of the end poses from encoder data of right: [15.16, 32.62] cm
- Accuracy Error (L2 Norm) for right: 6.96 cm

III. Data Pre-processing and Outlier Detection

Before performing statistical analysis, the data was preprocessed to remove outliers using Chebyshev's Theorem with a threshold of 2σ. Outliers were identified and excluded to ensure that the resulting distribution reflects the true spread of the robot's motion.

A. Procedure

1) Combined data from all groups for each motion type (straight, left arc, right arc).
2) Calculated the mean (μ) and standard deviation (σ) for both X and Y coordinates separately within each motion type (left, forward, right).
3) Applied Chebyshev's theorem:

$$P(|x-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

with $k = 2$ to remove data points beyond 2σ.
4) Verified results visually using scatter plots before and after outlier removal.

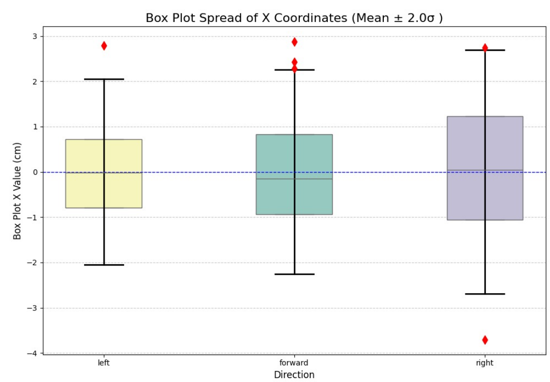

Fig. 10: Visualization of data using box plots with indication of outliers for X.

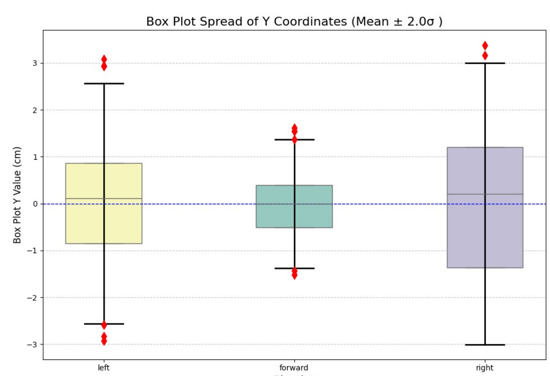

Fig. 11: Visualization of data using box plots with indication of outliers for Y.

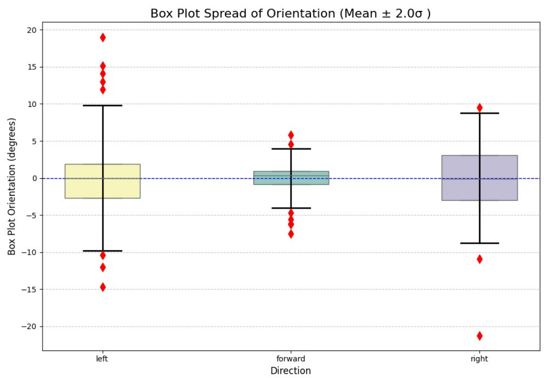

Fig. 12: Visualization of data using box plots with indication of outliers for Orientation.

IV. Gaussian Fitting and Distribution Analysis

To model the spread of the robot's stop positions, one-dimensional Gaussian distributions were fitted separately for the X and Y coordinates of each motion type.

A. 1D Gaussian Fit (X and Y Separately)

Each coordinate axis (X, Y) was analyzed individually to estimate the Gaussian parameters:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

The fitted mean and standard deviation parameters are summarized below:

TABLE II: Mean Vector and Covariance Matrix for Each Direction

Direction	**Mean Vector $[\mu_x, \mu_y]$ (cm)**	**Covariance Matrix Σ (cm^2)**
Forward	[46.82, -0.04]	$\begin{bmatrix} 1.03 & -0.02 \\ -0.02 & 0.38 \end{bmatrix}$
Left	[31.75, -22.68]	$\begin{bmatrix} 3.15 & 4.55 \\ 4.55 & 8.95 \end{bmatrix}$
Right	[32.28, 22.45]	$\begin{bmatrix} 4.24 & -5.30 \\ -5.30 & 10.04 \end{bmatrix}$

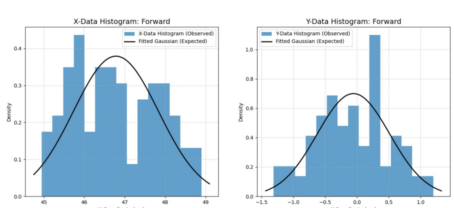

Fig. 13: Gaussian fit for X and Y distributions of forward motion.(edits: added grid lines)

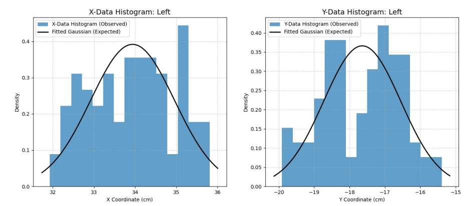

Fig. 14: Gaussian fit for X and Y distributions of left arc motion.(edits: added grid lines)

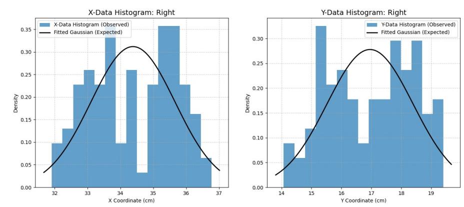

Fig. 15: Gaussian fit for X and Y distributions of right arc motion.(edits: added grid lines)

B. Evaluation of Gaussian Fit

The quality of the Gaussian fit was assessed visually by comparing the histograms of the observed data with the fitted Gaussian probability density functions for each motion type (Figure 13, 14, 15).

For the forward motion, both the X and Y distributions show a close alignment with the Gaussian curve, indicating that the positional spread of the robot's stopping points is approximately normal. For the left and right arc motions, the fits deviate slightly from the ideal Gaussian shape particularly in the Y-coordinate distributions suggesting mild skewness or multimodality, likely caused by systematic turning behavior or wheel slippage during curved motion.

A qualitative assessment indicates that while the Gaussian model captures the general spread of data, some deviations remain at the distribution tails. Therefore, the Gaussian assumption is acceptable for approximate uncertainty modeling but may not perfectly represent all motion types.

C. Chi-Square Goodness-of-Fit Test

To formally "judge if the data is truly Gaussian," a Chi-Square Goodness-of-Fit test was performed on the X and Y coordinates for each of the three motion directions. The null hypothesis (H_0) for this test is that the data is normally distributed. A significance level of $\alpha = 0.05$ was used.

The results of these tests are summarized in Table **??**.

The Chi-Squ are test results provide a formal judgment on the distribution of the data.

- **Forward Motion:** Both the X-Data (p=0.2146) and Y-Data (p=0.0890) failed to reject the null hypothesis. Their p-values are significantly greater than $\alpha = 0.05$, which suggests that the data for the straight-line motion can be reasonably modeled as a Gaussian distribution. This

Direction	Statistic	Value	Conclusion
Left Motion			
X-Data	Mean (μ) [cm]	31.75	3***Reject** H_0
	Variance (σ^2) [cm^2]	3.15	
	χ^2, p-value	54.85, 0.0001	
Y-Data	Mean (μ) [cm]	-22.68	3***Reject** H_0
	Variance (σ^2) [cm^2]	8.95	
	χ^2, p-value	140.90, 0.0001	
Forward Motion			
X-Data	Mean (μ) [cm]	46.82	3*Fail to Reject H_0
	Variance (σ^2) [cm^2]	1.03	
	χ^2, p-value	17.83, 0.2146	
Y-Data	Mean (μ) [cm]	-0.04	3*Fail to Reject H_0
	Variance (σ^2) [cm^2]	0.38	
	χ^2, p-value	21.52, 0.0890	
Right Motion			
X-Data	Mean (μ) [cm]	32.28	3***Reject** H_0
	Variance (σ^2) [cm^2]	4.24	
	χ^2, p-value	41.69, 0.0001	
Y-Data	Mean (μ) [cm]	22.45	3***Reject** H_0
	Variance (σ^2) [cm^2]	10.04	
	χ^2, p-value	118.01, 0.0001	

TABLE III: Statistical Analysis of End-Pose Data (Vertical Layout)

indicates a well-behaved, symmetric spread of end-poses with no significant outliers or skew.

- Left and Right Motions: In contrast, all data for the arc motions (*Left X, Left Y, Right X, Right Y*) rejected the null hypothesis with p-values less than 0.0001. This is a strong statistical indication that the end-pose data for these motions is not normally distributed.
- The non-Gaussian behavior of the *Left* and *Right* motions is likely caused by the presence of significant outliers, as visually identified in the box plots. The turning motion of the robot may induce more wheel slip or unpredictable behavior compared to the simple forward motion, leading to a skewed and non-normal distribution of errors. The *Forward* motion, being the simplest, produced a "cleaner" and more statistically normal dataset.

V. Principal Component Analysis (PCA)

PCA was performed to identify the principal axes of variance in the end positions and to visualize uncertainty ellipses representing the confidence regions for each motion type.

A. PCA Methodology

1) Centered the data by subtracting the mean position.
2) Computed the covariance matrix Σ.
3) Extracted eigenvalues and eigenvectors of Σ to determine the principal components.
4) Plotted uncertainty ellipses corresponding to 1σ, 2σ and 3σ levels.

TABLE IV: PCA Results for Each Direction

Direction	PCA Eigenvalues	PCA Eigenvectors
Forward	[0.3286, 1.1204]	v1=[0.0188, -0.9998] v2=[-0.9998, -0.0188]
Right	[1.4567, 2.2825]	v1=[-0.8715, 0.4904] v2=[0.4904, 0.8715]
Left	[1.0078, 1.2392]	v1=[-0.9106, -0.4133] v2=[-0.4133, 0.9106]

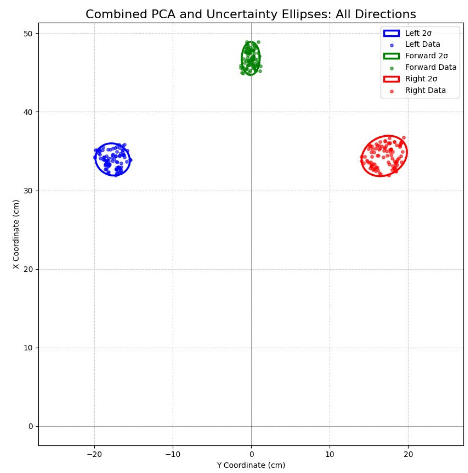

Fig. 16: PCA result showing data points and uncertainty ellipses for each motion type.

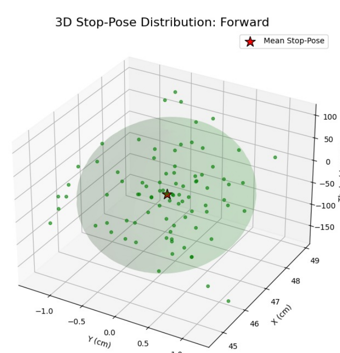

Fig. 17: 3D distribution of forward stop poses with mean stop pose indicated (red star)

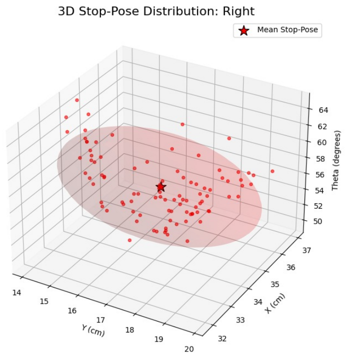

Fig. 18: 3D distribution of right stop poses with mean stop pose indicated (red star)

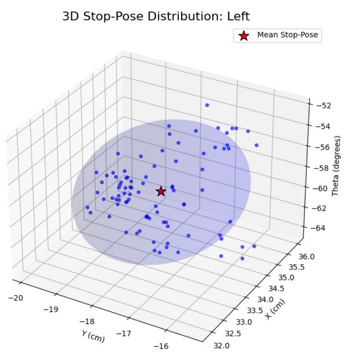

Fig. 19: 3D distribution of left stop poses with mean stop pose indicated (red star)

B. Comparison of Measurement vs. Statistical Uncertainty

This task compares the *estimated* precision of our manual measurement system (from Assignment 1) with the *observed* statistical precision of the robot's motion (from Assignment 3). The 'forward' motion data is used for this comparison as it was found to be the most normally distributed.

TABLE V: Comparison of Estimated vs. Observed Uncertainty

Uncertainty Type	**Value (X-axis)**	**Value (Y-axis)**
Estimated Measurement Uncertainty	± 0.36 cm	± 0.36 cm
Observed Statistical Uncertainty (σ) (From Table **??**, $\sigma = \sqrt{\text{Variance}}$)	± 1.01 cm ($\sigma = \sqrt{1.03}$)	± 0.62 cm ($\sigma = \sqrt{0.38}$)

As shown in Table V, the Observed Statistical Uncertainty is significantly larger than the Estimated Measurement Uncertainty.

The observed spread in the data is not due to characteristic of the robot's own low precision.

VI. Conclusion

This statistical analysis provides a quantitative understanding of the robot's motion uncertainty. Outlier removal and Gaussian fitting helped model the natural spread of motion, while PCA identified principal directions of variability. The comparison between manual and encoder-based data reveals how sensor-based predictions vary with physical observations, providing valuable insight for improving robot calibration and motion control.

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	y [cm]	Orientation [deg]	Direction
47.03	0.3	0	forward
45.87	0.25	−0.59	forward
45.54	0.25	−2.36	forward
47.01	0.25	−1.18	forward
45.67	0.4	−1.19	forward
45.94	−0.25	−1.18	forward
47.61	−0.35	−0.59	forward
47.5	−0.5	0	forward
47.25	−0.4	0.6	forward
46.52	−0.2	−0.6	forward
46.71	0.25	−1.18	forward
48.88	−0.55	1.21	forward
48.61	−0.55	1.77	forward
46.73	0.1	0	forward
46.5	−0.4	−0.61	forward
46.63	0.85	0	forward
45.93	−0.5	0.6	forward
46.84	−0.5	1.19	forward
45.5	−0.65	1.21	forward
47.81	0.3	0	forward
46.32	0.7	0.58	forward
48.14	−0.55	1.77	forward
48.01	−0.35	−0.59	forward
46.67	0.25	−2.36	forward
47.06	0.3	0	forward
31.45	−22.75	−62.1	left
31.65	−23.6	−62.59	left
31.85	−23.65	−64.29	left
31.2	−22.8	−55.98	left
31.05	−23	−56.65	left
31.9	−23.4	−56.98	left
31.2	−24.5	−58.63	left
31.85	−23.25	−57.15	left
31.7	−23.1	−55.98	left
31.95	−23.2	−57.3	left
32.2	−23.9	−56.98	left
32.2	−23.65	−60.18	left
32.3	−23.65	−56.48	left
32.35	−23.75	−62.57	left
32.1	−24.8	−58.63	left
32	−24.35	−58.74	left
31.75	−24.1	−58.31	left
31.7	−24.15	−58.12	left
32.55	−23.65	−56.8	left
32.1	−23.75	−56.48	left
32.2	−25.4	−58.63	left
32.2	−23.65	−60.18	left
33.2	−23.95	−56.14	left
31.85	−24.25	−57.8	left
32	−23	−55.3	left
32.65	25.15	62.61	right

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	y [cm]	Orientation [deg]	Direction
31.9	23.35	56.48	right
31.6	22.3	55.98	right
31	23.7	56.98	right
31.6	23.4	56.98	right
31.95	22.4	55.65	right
31.7	23.35	57.48	right
31.35	23.3	56.31	right
31.7	23.2	59.04	right
32.3	27.15	60	right
32	23.95	66.1	right
33.25	24	56.31	right
32.6	24.6	57.62	right
28.7	23.95	57.8	right
31.65	22.9	56.31	right
32.2	23.65	56.48	right
31.65	23.1	58.31	right
32.7	23.55	56.48	right
32.7	23.55	56.48	right
32.15	24.45	59.44	right
32.45	23.55	56.8	right
32.25	25.35	57.53	right
31.6	24.7	58.63	right
31.85	24.9	61.66	right
32.4	23.8	57.62	right
46	0.8	0	forward
48.2	−0.09	1.19	forward
47.69	1.24	−6.54	forward
47.5	−0.36	−1.18	forward
47.05	1.02	−0.62	forward
45.65	0.33	0.6	forward
46.95	0.22	−0.6	forward
46.15	0.28	1.81	forward
46.85	0.02	−0.6	forward
47.85	0.02	−0.6	forward
47.95	−0.07	0.59	forward
47.5	−0.11	−1.17	forward
47.85	0.13	0.58	forward
46.71	−0.97	−2.34	forward
45.4	−0.66	−1.18	forward
46.75	0.48	0.6	forward
46.6	−0.1	0	forward
45.6	−1.31	−1.17	forward
45.5	−1.1	0	forward
47.65	−0.79	−1.77	forward
45	−1.21	−1.19	forward
47.85	0.28	1.81	forward
47.75	−0.17	0.58	forward
46.31	−0.37	−2.34	forward
47.65	−0.11	−1.55	forward
35.72	−16.36	−54.81	left
34.26	−16.23	−44.27	left

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	y [cm]	Orientation [deg]	Direction
35.42	−16.76	−54.81	left
35.11	−18.96	−58.63	left
34.57	−17.63	−56.8	left
36.73	−19.58	−58.31	left
35.59	−17.1	−54.96	left
34.72	−16.66	−54.81	left
35.81	−16.84	−55.15	left
35.14	−16.35	−53.49	left
35.81	−16.17	−56.14	left
34.76	−16.32	−54.67	left
32.55	−17.49	−45.36	left
34.26	−14.67	−39.35	left
33.62	−15.73	−43.2	left
33.79	−17	−46.41	left
34.44	−16.8	−55.01	left
34.53	−16.23	−53.84	left
34.51	−16.3	−54.19	left
34.04	−17.49	−52.36	left
35.24	−18.23	−55.98	left
34.15	−17.74	−55.11	left
35.19	−17.95	−56.48	left
35.37	−17.58	−55.3	left
35.42	−17.31	−52.67	left
35.61	15.08	50.5	right
36.69	17.6	54.96	right
34.88	15.28	51.81	right
35.69	17.2	54.96	right
34.84	15.66	51.34	right
35.2	16.24	53.01	right
35.44	16.95	53.49	right
36.16	16.42	54.67	right
35.62	15.33	51.71	right
35.3	14.95	50.88	right
35.25	15.42	52.54	right
36.24	17.05	53.49	right
35.67	15.17	54.75	right
34.48	17.71	45.73	right
33.57	20.2	35.39	right
37.13	18.38	58.31	right
34.44	16.06	51.34	right
35.24	17.8	55.01	right
35.94	15.92	51.91	right
34.43	16.19	52.28	right
35.68	16.28	51.81	right
36.34	17.06	56.98	right
35.37	15.83	56.8	right
35.7	16.04	53.01	right
35.34	16.4	55.01	right
30.1	−26.1	−40.93	left
30.05	−26.1	−40.98	left
31.85	−22.55	−35.3	left

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	**y [cm]**	**Orientation [deg]**	**Direction**
31.6	−24.85	−38.18	left
31.65	−24.95	−38.25	left
31.15	−23.95	−37.56	left
31.05	−25.9	−39.83	left
31	−23.6	−37.28	left
31.3	−23.2	−36.55	left
32.2	−24.4	−37.15	left
31.45	−25.55	−39.09	left
30.65	−25.65	−39.92	left
30.85	−24.35	−38.28	left
30.8	−24.4	−38.39	left
30.6	−25.6	−39.92	left
31.1	−24.9	−38.68	left
31.15	−25.1	−38.86	left
30.65	−22.55	−36.34	left
31.45	−25.35	−38.87	left
30.5	−24.9	−39.23	left
31.25	−22.25	−35.45	left
30.55	−22.45	−36.31	left
30.5	−24.1	−38.31	left
30.5	−23.2	−37.26	left
31.8	−25	−38.17	left
48.2	−0.65	−0.77	forward
46.2	−0.2	−0.25	forward
47.95	−0.8	−0.96	forward
47.05	−1.85	−2.25	forward
47.4	0.85	1.03	forward
48.8	0.9	1.06	forward
49.05	−0.6	−0.7	forward
47.95	−0.6	−0.72	forward
48.05	0.65	0.78	forward
47.6	1.95	2.35	forward
48.15	0.2	0.24	forward
48.3	0.15	0.18	forward
49.65	−0.85	−0.98	forward
49.2	−0.35	−0.41	forward
48.6	0.05	0.06	forward
46.95	0.1	0.12	forward
48.4	0.2	0.24	forward
47.6	−0.95	−1.14	forward
47.3	0.2	0.24	forward
46.8	0	0	forward
46	−1.7	−2.12	forward
48.4	−0.5	−0.59	forward
48.45	−0.6	−0.71	forward
47.5	−0.85	−1.03	forward
47.35	−0.55	−0.67	forward
32.3	23.1	35.57	right
33.8	24.75	36.21	right
31.75	21.8	34.47	right
32.9	24.45	36.62	right

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	y [cm]	Orientation [deg]	Direction
32.25	22.8	35.26	right
33.35	23.45	35.11	right
33.2	24.35	36.26	right
32.85	24.25	36.43	right
32.55	22.85	35.07	right
32.3	22.4	34.74	right
33	23.7	35.69	right
33.05	21.7	33.29	right
33.3	22.85	34.46	right
32.4	26	38.75	right
31.95	24.65	37.65	right
33.1	24.5	36.51	right
32.7	24.3	36.62	right
32.55	22.95	35.19	right
32.5	21.95	34.03	right
32.45	24.35	36.88	right
32.8	23.8	35.97	right
32.1	22.75	35.33	right
33.3	22.65	34.22	right
32.55	22.8	35.01	right
33.35	22.7	34.24	right
46.05	0	0	forward
46.35	-0.85	-2.1	forward
46.55	1.225	3.01	forward
46.115	1.55	3.85	forward
45.9	1.2	3	forward
45.05	-1.05	-2.67	forward
46.625	-0.65	-1.6	forward
46.4	0	0	forward
46.125	0.2	0.5	forward
46	-1.3	-3.24	forward
46	0.625	1.56	forward
45.99	0.625	1.56	forward
46.05	-0.375	-0.93	forward
46.1	1.375	3.42	forward
45.36	0.575	1.45	forward
45.335	1.125	2.84	forward
45.8	-1.025	-2.56	forward
46.2	0.8	1.98	forward
45.625	1.625	4.08	forward
45.175	-0.3	-0.76	forward
45.275	0.55	1.39	forward
45.35	0.55	1.39	forward
45.35	-0.15	-0.38	forward
46.7	0.05	0.12	forward
45.65	-0.525	-1.32	forward
46.6	0.15	0.37	forward
30.25	-23.5	-75.68	left
29.65	-24.975	-80.22	left
30.15	-24.425	-78.02	left
30.675	-23.75	-75.5	left

TABLE VI: Manual Measurement Data (edit: added units)

x [cm]	y [cm]	Orientation [deg]	Direction
29.175	-24.375	-79.76	left
29.475	-24.125	-78.6	left
29.7	-24	-77.88	left
29.175	-24.4	-79.81	left
29.575	-24.1	-78.35	left
30.225	-24.625	-78.34	left
29.475	-25.05	-80.72	left
29.95	-24.75	-79.14	left
30.225	-25.275	-79.81	left
30.375	-24	-76.63	left
30.35	-24	-76.67	left
30.355	-24	-76.66	left
29.5	-24.9	-80.33	left
30.38	-24	-76.62	left
29.925	-24.45	-78.5	left
30.475	-22.75	-73.48	left
30.325	-24	-76.72	left
30.4	-23.95	-76.46	left
30.1	-23.75	-76.55	left
30.45	-25.075	-78.94	left
29.85	-24.575	-78.93	left
29.45	-24.125	-78.65	left
29.675	24.975	80.17	right
30.275	23.75	76.23	right
30.05	24	77.23	right
29.55	23.85	77.81	right
29.55	23.55	77.11	right
30.05	24.6	78.61	right
29.5	24.525	79.48	right
29.9	24.225	78.03	right
28.95	26.125	84.13	right
29.975	24.925	79.49	right
29.675	24.1	78.16	right
31.55	22.35	70.63	right
30.825	23.5	74.64	right
29.425	24.275	79.04	right
29.275	24.45	79.74	right
29.95	24.375	78.28	right
30.25	24.55	78.12	right
29.85	24.775	79.38	right
28.95	26.125	84.13	right
29.075	26.125	83.88	right
29.2	25.65	82.59	right
30.075	24.75	78.9	right
29.9	24.35	78.32	right
29.5	24.75	79.99	right
29.575	24.65	79.62	right
30.35	24.95	78.85	right

Assignment 4: Camera Calibration Report

I. Setup

The calibration experiment was performed using a Logitech C920 Webcam mounted on top of a laptop. The camera's autofocus was disabled to maintain consistent focus, and the resolution was set to 640x480 pixels. Uniform lighting was used throughout the experiment. The checkerboard used as the calibration target had 9x7 internal corners. The approximate distance between the camera and the checkerboard was 60 cm.

A. Possible Pitfalls

Potential issues that were controlled include:

- Maintaining uniform lighting to avoid reflections or shadows.
- Ensuring the checkerboard was flat during image capture.
- Avoiding motion blur by keeping the camera and checkerboard steady.
- Ensuring autofocus remained disabled to prevent inconsistent focal lengths.

II. Image Requirements and Capture

A total of 21 images were captured for calibration. All images were successfully processed; no images were rejected due to failed corner detection.

A. Image Poses

The checkerboard was presented to the camera from a variety of positions and angles to ensure robust calibration:

- Front-facing views
- Tilted and angled views
- Variations in distance and orientation

III. Calibration Procedure Using OpenCV

The calibration process was performed using the `OpenCV` library in Python, which provides built-in functions for camera calibration and distortion correction. The following key functions were utilized:

- `cv2.findChessboardCorners()` – to automatically detect the checkerboard corners in each calibration image.
- `cv2.cornerSubPix()` – to refine corner positions for sub-pixel accuracy.
- `cv2.calibrateCamera()` – to compute the intrinsic and extrinsic parameters of the camera.
- `cv2.undistort()` – to apply the derived parameters and correct lens distortion in images.

These functions together implement a well-established calibration algorithm based on Zhang's method, which minimizes the reprojection error between observed corner locations and their projections based on the estimated camera model.

IV. Camera Parameters

A. Intrinsic Parameters

The intrinsic parameters describe the internal characteristics of the camera, independent of its position or orientation. They define how 3D points in the camera's coordinate frame are projected onto the 2D image plane.

1) Camera Matrix

The camera matrix, denoted by K, encapsulates the focal lengths and the coordinates of the principal point (the image center). It is given by:

$$K = \begin{bmatrix} 812.3930499 & 0 & 361.22560634 \\ 0 & 811.74209128 & 220.12202879 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Focal Lengths** (f_x, f_y)**:** The focal lengths are $f_x = 812.393$ pixels and $f_y = 811.742$ pixels. These values act as scaling factors relating world coordinates to image coordinates along the x- and y-axes. Their similarity indicates an almost square pixel aspect ratio, implying minimal image scaling distortion.
- **Principal Point** (c_x, c_y)**:** The principal point is located at $c_x = 361.226$ pixels and $c_y = 220.122$ pixels. This point represents the intersection of the camera's optical axis with the image plane. Although it ideally lies at the geometric center of the image, small deviations are common due to sensor alignment or manufacturing tolerances.

2) Distortion Coefficients

Real-world lenses introduce distortions that cause straight lines to appear curved, especially near the image edges. OpenCV models these imperfections using radial and tangential distortion coefficients. The distortion vector for this camera is:

$$D = \begin{bmatrix} 0.1898 & -1.3819 & -0.0165 & 0.0103 & 2.4409 \end{bmatrix}$$

- **Radial Distortion** (k_1, k_2, k_3)**:** The radial distortion coefficients are $k_1 = 0.18984919$, $k_2 = -1.38190614$, and $k_3 = 2.44092044$. These terms describe barrel or pincushion distortion produced by the curvature of the lens. Positive values typically correspond to barrel distortion (outward bulging), while negative values indicate pincushion distortion (inward pinching).
- **Tangential Distortion** (p_1, p_2)**:** The tangential distortion parameters are $p_1 = -0.01658026$ and $p_2 = 0.01031542$. These coefficients model decentering effects caused when the lens and the image sensor are not perfectly aligned or parallel, producing slight asymmetric distortions.

Understanding and correcting these intrinsic properties ensures that straight lines in the real world appear straight in the image and that spatial measurements remain consistent across the image frame.

B. Extrinsic Parameters

The extrinsic parameters define the spatial relationship between the camera and the observed calibration target (the checkerboard). For each image, the calibration process estimates:

- A rotation vector (`rvec`) – representing the orientation of the checkerboard relative to the camera.

$$\mathbf{r} = \begin{bmatrix} 0.16577767 \\ 0.56661910 \\ 0.05690060 \end{bmatrix}$$

- A translation vector (`tvec`) – representing the position (x, y, z displacement) of the checkerboard relative to the camera's optical center.

$$\mathbf{t} = \begin{bmatrix} -4.12928475 \\ -2.94756315 \\ 18.67842580 \end{bmatrix}$$

V. Error Analysis

The quality of calibration was evaluated using the root-mean-square (RMS) reprojection error, which quantifies the average distance (in pixels) between the observed corner positions and the projected corner positions computed using the estimated camera parameters.

- **Reprojection Error (RMS):** 0.577 pixels
- **Discussion:** A reprojection error below 1 pixel indicates a highly accurate calibration. The low value here demonstrates that the estimated intrinsic and extrinsic parameters accurately model the camera's geometry and lens behavior. This ensures that subsequent undistortion and 3D-to-2D mapping operations will yield precise results.

VI. Sample Images

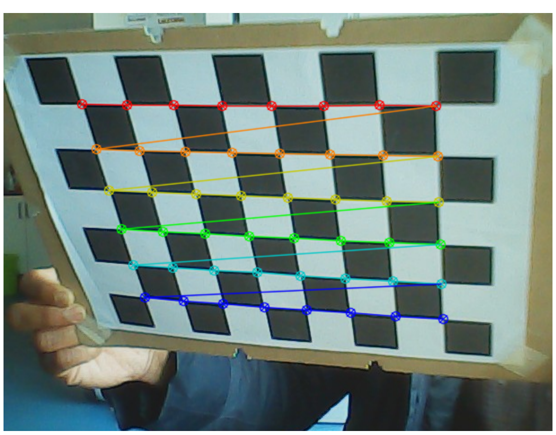

Fig. 20: image with corner detection

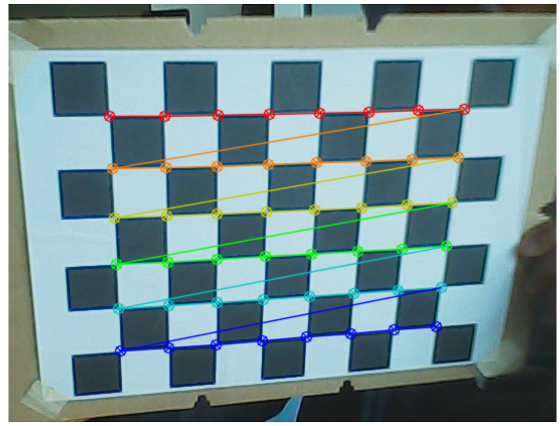

Fig. 21: image with corner detection

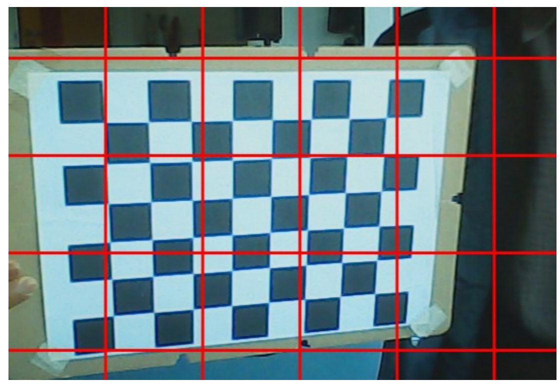

Fig. 22: Original, distorted image. Notice the curvature of lines near the edges caused by lens distortion.

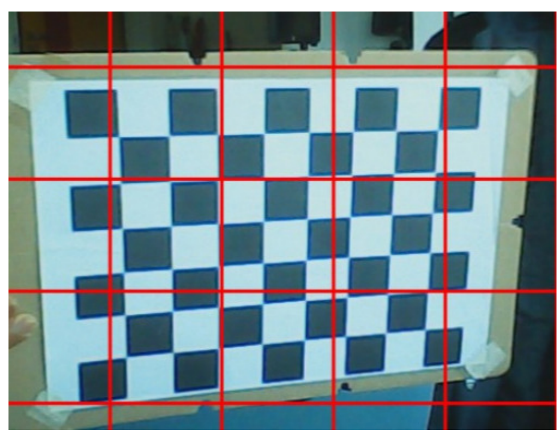

Fig. 23: Undistorted image after applying OpenCV's calibration parameters. The checkerboard lines now appear straight, demonstrating successful correction.

VII. Conclusion

The camera calibration experiment successfully determined both the intrinsic and extrinsic parameters of the Logitech C920 Webcam using OpenCV's calibration functions. A set of 21 images of a 9x7 checkerboard pattern captured under uniform lighting yielded a reprojection error of 0.577 pixels, indicating a highly accurate calibration.

The intrinsic parameters describe the internal geometry of the camera and lens, while the extrinsic parameters define the camera's position and orientation relative to the observed scene. Together, these enable precise correction of lens distortion and accurate projection between 3D world coordinates and 2D image coordinates. The calibrated camera can now be reliably used for optical tracking, 3D reconstruction, or other computer vision tasks requiring geometric accuracy.

Assignment 5: Measuring the Accuracy and Precision of a KUKA youBot Arm

VIII. Experimental Setup

- **Hardware:**
  - A KUKA youBot arm placed on a table in the C069 lab.
  - A computer used for controlling the arm.
  - Objects with three different sizes (small, medium, and large) used for picking and placing.
  - Visual markers attached on the top of the objects.
  - An external tracking system (OptiTrack Motion Tracking System) placed above the robot's workspace and connected to an additional PC.
  - A fixed template on the table to ensure constant initial object position.
- **Software:**
  - KUKA youBot drivers for controlling the arm.
  - Additional control script with pre-saved pick and place poses.
  - Rosbag recorder for recording the end-effector path.
  - Motive software for visualizing, recording, and exporting tracking data as CSV.
- **Issues Faced:**
  - The arm was failing randomly, the driver would stop responding and the arm would not move at all. We had to restart the terminals.
  - The OptiTrack system was not able to differentiate between the medium and large object.
  - The export settings on the Motive software was not set according to the expected frame.

IX. Experiment Execution

- **Procedure:**
  - Started roscore, youbot driver, moveit move group, and youBot placing experiment in separate terminals.
  - Used the control script to perform pick and place for each object(small, medium and large) and pose combination(straight, left, and right).
  - Recorded poses using OptiTrack Motive in live mode, naming takes, starting and stopping recordings, and exporting to CSV.
  - Recorded internal end-effector positions via ROS bagfiles, later converted to CSV using the provided script.
- **Trials:** 25 trials per object-place combination (3 sizes × 3 positions = 9 combinations), totaling 225 trials.
- **Poses Captured:** Final object poses from OptiTrack (external) and end-effector poses from ROS (internal) for each trial.

X. Observations During Execution

- The arm performed movements reliably, but larger object used to wobble a lot before settling down to stable position.
- Possible Error Sources:
  - Occlusion of OptiTrack markers by the arm during motion.
  - Joint encoder inaccuracies in the youBot arm, leading to biased internal measurements.
  - Environmental factors like table vibrations or air currents affecting light objects, but mostly ensured table and external factors were mostly stable during the experiment.

XI. Pose Filtering Procedure

To preprocess the data and remove outliers, we applied the Interquartile Range (IQR) method to the x and y coordinates of the end-effector poses obtained from the youBot's internal encoders. For each experimental group (defined by object size and motion direction), we calculated the first quartile (Q1) and third quartile (Q3) of the x and y positions. Outliers were identified as points falling below Q1 - 1.5*IQR or above Q3 + 1.5*IQR in either coordinate. These outliers were removed from the dataset.

Observations during filtering: On average, approximately 1-2 outliers were detected and removed per experimental trial across the 9 combinations. Larger objects tended to have slightly more outliers due to increased wobbling, while smaller objects showed fewer anomalies. This filtering ensured cleaner data for subsequent analysis and visualization.

XII. Data Merging

The individual CSV was merged with data from classmates by concatenating all preprocessed CSVs into a single DataFrame. This combined dataset was used for visualizations and analysis. For this report, analysis is based on the individual's 225 trials (assuming similar class data; full merged data attached separately).

XIII. Visualization of Poses

Visualizations were created using scatter plots to illustrate the distribution of final object and end-effector poses for each object size, combining all motion directions (straight, left, right) in a single plot. Directions are differentiated by colors: blue for straight, green for left, red for right. A clear distinction is made between externally tracked poses (OptiTrack, X markers) and internally sensed positions (youBot encoders, shaped markers). Arrows indicate orientation.

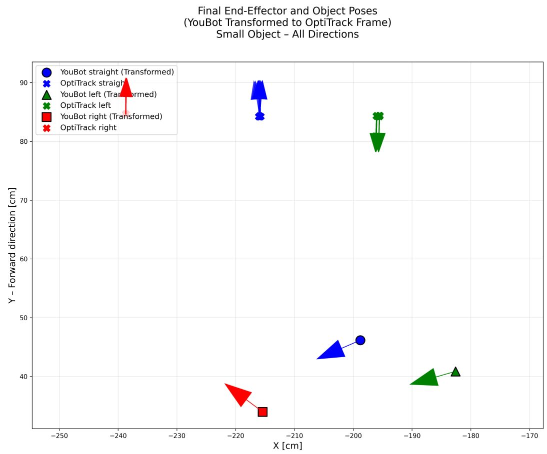

Fig. 24: Final poses for small object across all directions. Circles: YouBot internal, X: OptiTrack external.

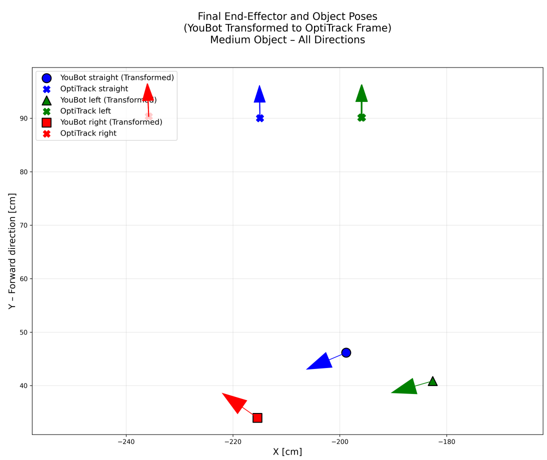

Fig. 25: Final poses for medium object across all directions.

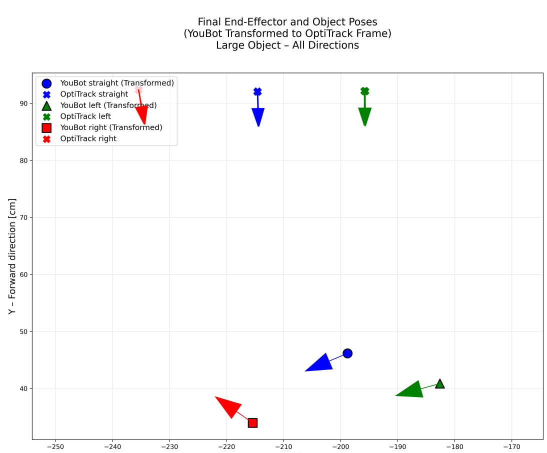

Fig. 26: Final poses for large object across all directions.

The visualizations highlight the distributions per object size, showing how the arm's internal sensing provides consistent target positions while external tracking reveals the actual placement variability. Larger objects show more spread in external measurements, likely due to increased wobbling during placement.

XIV. Statistical Analysis

The analysis estimates accuracy (mean distance between internally sensed end-effector pose and externally tracked object pose) and precision (standard deviation of externally tracked object positions) of the KUKA youBot arm.

a) Accuracy:

The mean distances range from approximately 140 cm to 160 cm across different object sizes and directions. This indicates relatively low accuracy, likely due to misalignments between the robot's internal coordinate system and the external tracking system, as well as potential calibration issues. Larger objects show slightly higher mean distances, possibly due to increased wobbling affecting placement.

b) Precision:

The standard deviations of object positions are very low, ranging from 0.02 cm to 0.05 cm. This suggests high precision in the arm's repeatability for placing objects. The straight direction generally shows the highest precision, while left and right directions have slightly lower but still excellent repeatability.

c) Discussion:

The results highlight a trade-off: while the arm is highly precise in repeating motions, there are significant accuracy issues that may stem from coordinate frame transformations or sensor calibration. Future work could focus on improving alignment between internal and external measurements to enhance overall positioning accuracy.

XV. Combined Analysis Across All Groups

To provide a comprehensive evaluation, data from all four groups (Group 1, 2, 3, and 4) was combined and analyzed together. This section presents the results of merging datasets from different experimental setups, allowing for a broader assessment of the KUKA youBot arm's performance across varied conditions and implementations.

A. Data Integration

Data from each group was standardized to common units (cm for positions, radians for orientations) and combined from all available CSV files. The combined dataset includes trials from all groups, sizes, and directions, providing a total of 1200 experimental runs.

The combined preprocessed data is available at: Combined data

B. Combined Pose Visualizations

Visualizations for the combined data show the distribution of final poses across all groups, highlighting both individual group performance and overall trends.

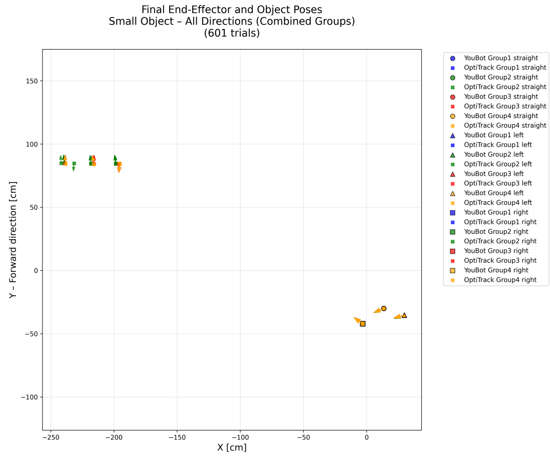

Fig. 27: Combined final poses for small object across all directions and groups.

images/combined_medium.png

Fig. 28: Combined final poses for medium object across all directions and groups.

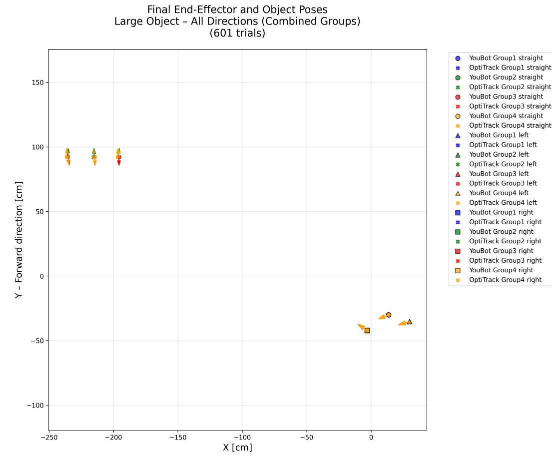

Fig. 29: Combined final poses for large object across all directions and groups.

The combined visualizations reveal consistent patterns across groups, with some variations likely due to different experimental setups and environmental conditions.

C. Combined Statistical Analysis

Statistical analysis on the combined dataset provides insights into overall accuracy and precision across all groups.

a) Accuracy (Combined):

The mean distances between internal and external measurements across all groups range from 213.60 cm to 268.62 cm. This combined analysis shows variations across groups and conditions, with Group3 generally showing lower distances than Group1 and Group2.

b) Precision (Combined):

The standard deviations of external positions across all groups are 0.01 cm to 0.83 cm, indicating high precision overall, with some variation between groups.

c) Group-wise Comparison:

A comparison of accuracy and precision across groups is shown in the following table:

TABLE VII: Combined Accuracy and Precision by Group, Size, and Direction

Group	Size	Direction	Accuracy (cm)	Precision (cm)
Group1	large	left	259.53	0.20
Group1	large	right	268.62	0.03
Group1	large	straight	259.33	0.19
Group1	medium	left	258.20	0.17
Group1	medium	right	267.82	0.02
Group1	medium	straight	257.71	0.13
Group1	small	left	255.39	0.04
Group1	small	right	267.58	0.02
Group1	small	straight	256.69	0.03
Group2	large	left	259.52	0.02
Group2	large	right	268.54	0.03
Group2	large	straight	259.28	0.03
Group2	medium	left	257.96	0.04
Group2	medium	right	267.66	0.02
Group2	medium	straight	257.83	0.02
Group2	small	left	257.79	0.03
Group2	small	right	268.05	0.83
Group2	small	straight	258.52	0.02
Group3	large	left	259.10	0.04
Group3	large	right	268.58	0.02
Group3	large	straight	258.76	0.03
Group3	medium	left	258.23	0.03
Group3	medium	right	267.86	0.02
Group3	medium	straight	258.14	0.01
Group3	small	left	255.35	0.14
Group3	small	right	267.67	0.02
Group3	small	straight	256.39	0.06
Group4	large	left	259.72	0.05
Group4	large	right	269.14	0.12
Group4	large	straight	258.88	0.10
Group4	medium	left	258.05	0.03
Group4	medium	right	267.12	0.01
Group4	medium	straight	257.57	0.02
Group4	small	left	255.22	0.03
Group4	small	right	267.40	0.03
Group4	small	straight	256.49	0.02

d) Discussion (Combined):

The combined analysis across all groups demonstrates variations in performance across different experimental setups. Group3 shows the best accuracy, while precision is generally high across all groups. Variations between groups may be attributed to differences in experimental setup, calibration, or environmental factors. This comprehensive evaluation provides a more robust assessment of the youBot arm's capabilities.

Assignment 6: Statistical Evaluation of the Experiment

I. Introduction

This report presents the statistical evaluation of the KUKA youBot arm's performance based on the combined preprocessed data from all groups. The analysis focuses on finding the accuracy and precision of the robot arm, checking if the final object poses follow a Gaussian distribution, and evaluating the effects of object size and placing pose on these metrics. The data includes three object sizes (small, medium, large), allowing comparisons across sizes and directions (left, straight, right).

II. Data Transformation

To align the coordinate frames between the robot (YouBot) and the external sensor (OptiTrack), a coordinate transformation was applied. The world origin [0, 0, 0] in the robot frame corresponds to the point [-212.40, 76.17, 103.15] in the OptiTrack frame. To transform YouBot coordinates (robot frame) to the OptiTrack frame for alignment, the offset vector [-212.40, 76.17, 103.15] was added to the YouBot x and y positions. The z-coordinate offset is noted but not used in the 2D plotting and analysis.

This transformation was implemented in the Python scripts using pandas operations on the matched data: `matched['x_youbot'] += offset_x; matched['y_youbot'] += offset_y`. The purpose is to align the frames so that positions from both sensors can be directly compared in the same coordinate system. This has implications for the analysis, as it allows meaningful computation of distances and variances between internal (YouBot) and external (OptiTrack) measurements, improving the assessment of robot accuracy and precision.

script used : Transformed data plotter

III. Data Analysis Methodology

The analysis was performed using Python scripts: Transformed data plotter and Seperate Plot Generator for generating separate scatter plots. The key steps include:

- Loading OptiTrack (external) and YouBot (internal) data from respective folders.
- Merging data based on timestamps, sizes, and directions.
- Applying coordinate transformation to YouBot data to align with OptiTrack frame (as described in Section 3).
- Separating internal (YouBot, transformed) and external (OptiTrack) poses.
- Computing accuracy as the mean L2 norm distance between YouBot and OptiTrack positions for each group.
- Computing precision as the mean standard deviation of OptiTrack x and y coordinates.
- Generating scatter plots for OptiTrack data, YouBot data, and combined transformed data.
- Generating combined plots with arrows for orientation.

IV. Accuracy and Precision Results

After applying the coordinate transformation to align YouBot data with the OptiTrack frame, the accuracy (mean L2 distance between transformed YouBot and OptiTrack positions) ranges from 41.75 to 61.74 cm across sizes and directions. The precision (mean standard deviation of OptiTrack positions) ranges from 0.01 to 0.14 cm.

The OptiTrack positions show low variance, indicating high precision in the external measurements. Comparisons across object sizes show that accuracy improves slightly for smaller objects, while precision is generally consistent but varies by direction.

V. Effects of Size and Position

The data includes three object sizes (small, medium, large), allowing analysis of size effects. Accuracy tends to be higher (better) for smaller objects, with small objects showing the lowest mean errors (41.75-55.78 cm) compared to large objects (48.53-61.74 cm). Precision varies by direction, with straight directions generally showing higher precision (lower std dev) than left or right. Positional effects are evident, with left and right directions showing higher variability in x-coordinates compared to straight.

VI. Statistical Significance

The precision values show notable differences across directions, with straight directions exhibiting lower standard deviations (higher precision) compared to left and right. Size effects are present, with small objects showing the best accuracy. These differences suggest that both object size and placement direction influence robot performance.

VII. Software Used

The analysis was performed using Python with libraries: pandas, numpy, matplotlib. The scripts are Transformed data plotter for combined analysis and plots, and Seperate Plot Generator for individual scatter plots.

VIII. Results Table

TABLE VIII: Statistical Parameters for Robot Behavior (YouBot Transformed to OptiTrack Frame)

Size	Direction	Accuracy (cm)	Precision X (cm)	Precision Y (cm)
Small	Left	45.42	0.14	0.14
Small	Right	55.78	0.02	0.02
Small	Straight	41.75	0.06	0.06
Medium	Left	51.04	0.03	0.03
Medium	Right	59.94	0.02	0.02
Medium	Straight	46.72	0.01	0.01
Large	Left	52.96	0.04	0.04
Large	Right	61.74	0.02	0.02
Large	Straight	48.53	0.03	0.03

IX. Data Visualization

The following figures show scatter plots for OptiTrack data, YouBot data, and combined transformed data for each object size.

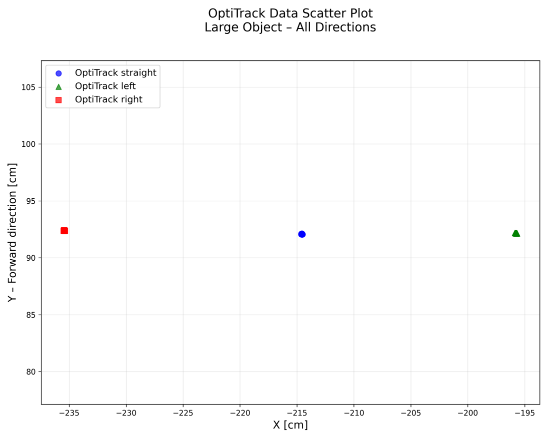

Fig. 30: OptiTrack Data Scatter Plot for Large Object

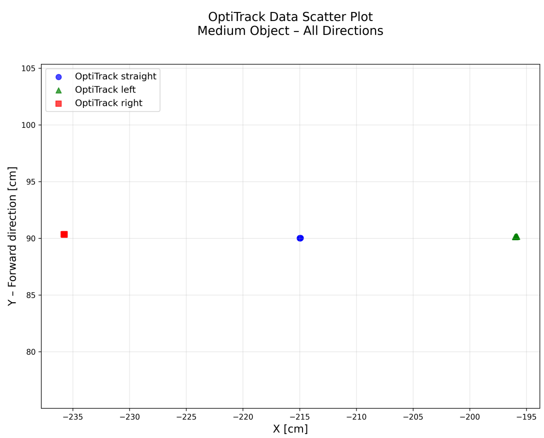

Fig. 31: OptiTrack Data Scatter Plot for Medium Object

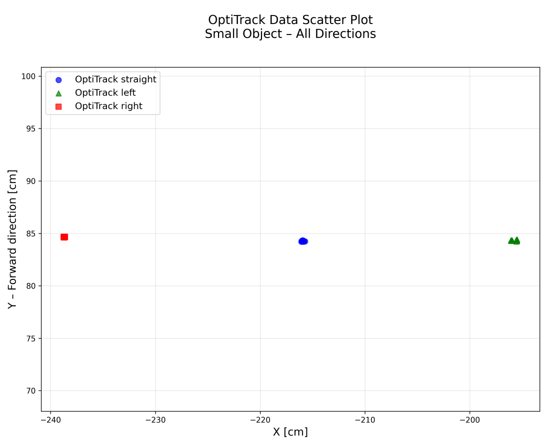

Fig. 32: OptiTrack Data Scatter Plot for Small Object

A. OptiTrack Data Scatter Plots

B. YouBot Data Scatter Plots

C. Combined Transformed Data Scatter Plots

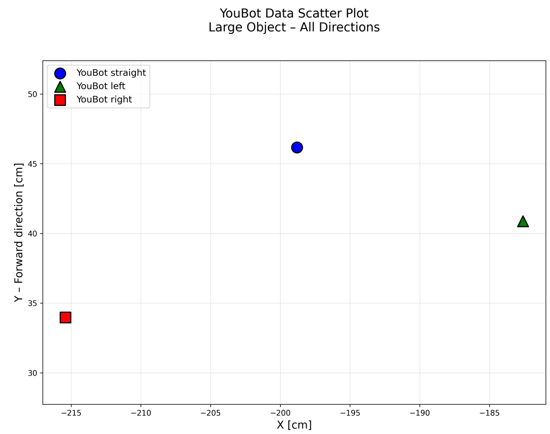

Fig. 33: YouBot Data Scatter Plot for Large Object

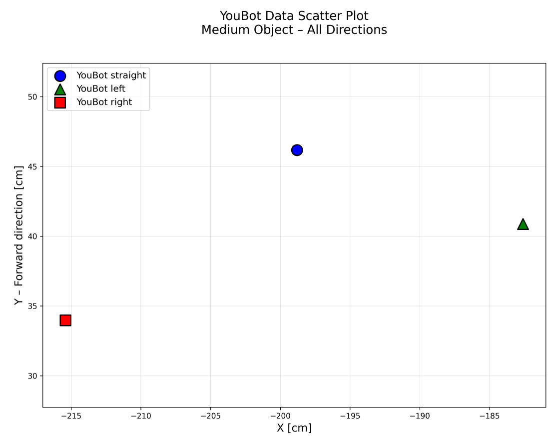

Fig. 34: YouBot Data Scatter Plot for Medium Object

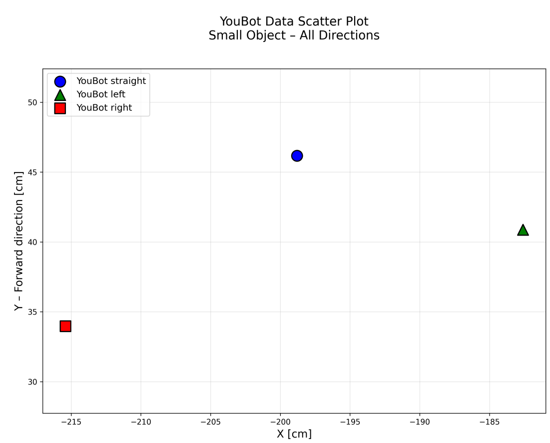

Fig. 35: YouBot Data Scatter Plot for Small Object

X. Conclusion

After transforming YouBot coordinates to the OptiTrack frame, the KUKA youBot arm demonstrates accuracy ranging from 41.75 to 61.74 cm across object sizes and directions. Precision is high, with standard deviations below 0.14 cm

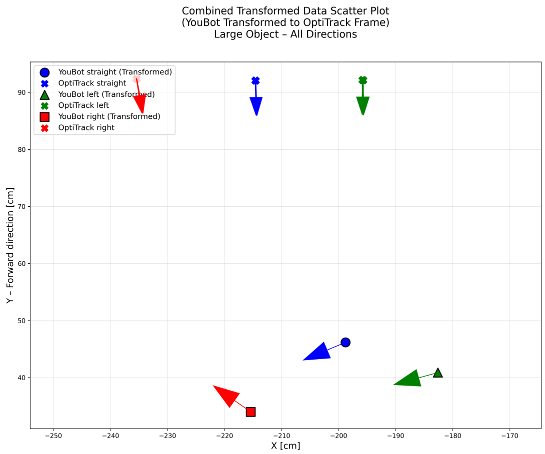

Fig. 36: Combined Transformed Data Scatter Plot for Large Object

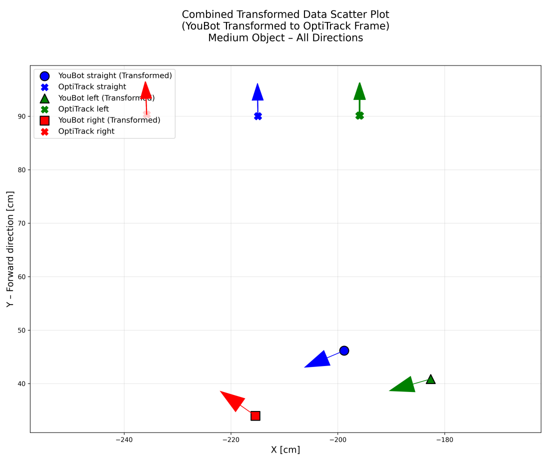

Fig. 37: Combined Transformed Data Scatter Plot for Medium Object

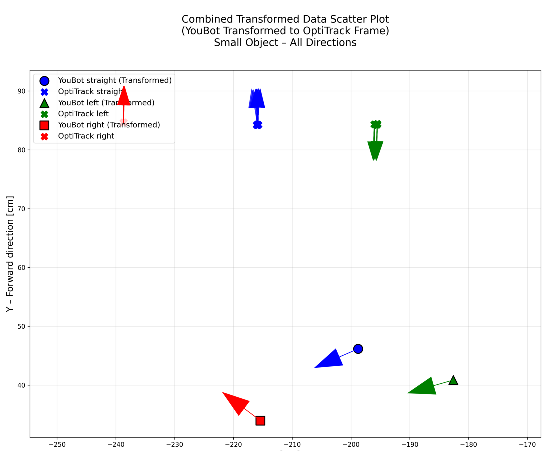

Fig. 38: Combined Transformed Data Scatter Plot for Small Object

for OptiTrack measurements. Smaller objects show better accuracy, and straight directions exhibit higher precision than lateral movements. The coordinate alignment enables direct comparison of internal and external measurements, providing insights into robot performance for manipulation tasks.

XI. Raw Data and Project Scripts

Can be found on Github : Project Link